

Classic Papers in Physical Oceanography
Instructor F. Straneo
List of Papers
Spring Semester - 2005

The Scientific Method

Week 1: Basic Theories

Chalmers, A.F., What is this thing called Science?, Univ. of Queensland Press.

Ed. Hackett.

Introduction: what is the scientific method?

Chapter 1: Science as knowledge derived from facts.

Chapter 2: Observation as practical intervention

Chapter 3: Experiment

Chapter 4: Induction

Chapter 5: Popper's Falsificationism

Thomas Kuhn: Scientific Revolutions

from *The Philosophy of Science*, pp. 139-157. Eds. Boyd, Gasper and Trout.

Medawar: Advice to a Young Scientist

Chapter 11: the scientific process (pp. 82-94).

Abyssal Circulation and MOC

Week 2: Thermohaline Circulation

Wunsch C., 2002: What is the thermohaline circulation? *Science*, 298, 1179-1180.

Rahmstorf, S., 2003: The current climate. *Nature*, 421, 699.

Stommel, H. 1961: Thermohaline convection with two stable regimes. *Tellus*,
13, 224-230.

Week 3: Theory for the Abyssal Circulation

Stommel, H., A.B. Arons, and A.J. Faller, 1958: Some of examples of stationary planetary flow patterns in bounded basins. *Tellus*, 10(2), 179-187.

Stommel, H. and A. B. Arons, 1960: On the abyssal circulation of the world ocean --- I. Stationary planetary flow patterns on a sphere, *Deep Sea-Res.*, 6, 140-

Week 4: Observing the Abyssal Circulation

Crease, J., 1962: Velocity Measurements in the Deep Water of the Western North Atlantic. *J. Geophys. Res.*, 67, 3173-3176.

Hogg, N.H. And W.B. Owens, 1999: Direct Measurements of the deep circulation within the Brazil Basin, *Deep Sea-Res. II*, 46, 335-353.

Week 5: Diapycnal Mixing

Polzin, K., J.M. Toole, J.R. Ledwell and R.W. Schmitt, 1997: Spatial Variability of Turbulent Mixing in the Abyssal Ocean . *Science*, 276, 93-96.

Munk, W. and C. Wunsch, 1998: Abyssal Recipes II: energetics of wind and tidal mixing. *Deep-Sea Res. I*, 45, 1977-2010.

Eddy Fluxes and Lateral Mixing

Week 6: Lateral Mixing in the Ocean

Iselin, C.O., 1939: The influence of vertical and lateral turbulence on the characteristics of the waters at mid-depths. *Eos Trans., AGU*, 20, 414-417.

Montgomery, R.B., 1940: The present evidence on the importance of lateral mixing processes in the ocean. *Bull. Americ. Meteor. Soc.*, 21, 87-94.

Gent P. and J.C. McWilliams, 1990: Isopycnal Mixing in Ocean Circulation Models. *J. Phys. Ocean.*, 20, 150-155.

Week 7: Lateral Mixing inferred from Lagrangian Measurements

Taylor, 1921: Diffusion by continuous movements. *Proc. Lond. Math. Soc.*, Ser. 2, 20, 196-211.

Freeland, Rhines and Rossby (1975): Statistical observations of the trajectories of neutrally buoyant floats in the North Atlantic. *J. Mar. Research.* 33, 383-404.

Week 8: Eddy-Mean Flow Interaction and EKE Distribution

Holland and Lin, 1975a: On the generation of Mesoscale Eddies and their Contribution of the Oceanic General Circulation. I. A preliminary Numerical Experiment., *J. Phys. Ocean.*, **5**, 642-657.

Stammer, D., 1997: Global Characteristics of Ocean Variability Estimated from Regional TOPEX/POSEIDON Altimeter Measurements. *J. Phys. Ocean.*, **27**, 1743-1769.

Munk, W. and C. Wunsch, 1998: Abyssal Recipes II: energetics of wind and tidal mixing. *Deep-Sea Res. I*, **45**, 1977-2010.

Wind-Driven Circulation

Week 9: Sverdrup Balance

Letmaa, Niiler and Stommel, 1977: Does the Sverdrup Relation account for the Mid-Atlantic Circulation? *J. Mar. Res.*, **35**, 1-10.

Wunsch, C. and Roemmich, D, 1985: Is the North Atlantic in Sverdrup Balance? *J. Phys. Ocean.*, **15**, 1876-1880.

Week 10: The Unventilated Thermocline

Rhines and Young, 1982: A theory of wind-driven circulation. I. Mid-Ocean Gyres. *J. Mar. Res.*, **40**, 559-596.

Week 11: The Ventilated Thermocline

Luyten, Pedlosky and Stommel, 1983: The Ventilated Thermocline. *J. Phys. Ocean.*, **13**, 292-309.